

MSTAR PRACTICE WORKSHEET**Lesson 4-1: Understand Percent**

Grade 6 Mathematics · Standard 6.AT.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: C**

What does the word percent mean?

- ☐ A Per ten
- ☐ B Per thousand
- ☒ C Per hundred
- ☐ D Per part

Percent means per hundred, which is why every percent can be written as a fraction with a denominator of 100.

Question 2 **SELECTED RESPONSE — CORRECT: D**

3 out of every 10 students walk to school. What percent is that?

- ☐ A 3%
- ☐ B 10%
- ☐ C 13%
- ☒ D 30%

Multiply both terms of the ratio by 10 to compare to 100: 3 out of 10 becomes 30 out of 100, or 30%.

Question 3 SELECTED RESPONSE — CORRECT: C

A decimal grid has 100 squares and 65 are shaded. What percent is shaded?

- ☐ A 6.5%
- ☐ B 35%
- ☒ C 65%
- ☐ D 100%

Each square is one percent, so 65 shaded squares is 65%.

Question 4 SELECTED RESPONSE — CORRECT: D

Which percent means MORE than one whole?

- ☐ A 0.8%
- ☐ B 80%
- ☐ C 100%
- ☒ D 120%

100% is exactly one whole, so any percent above 100 is more than one whole.

Question 5 SELECTED RESPONSE — CORRECT: D

The orange kitten weighs 1.5 times as much as the gray kitten. What percent of the gray kitten's weight is that?

- ☐ A 1.5%
- ☐ B 15%
- ☐ C 50%
- ☒ D 150%

The gray kitten is the whole, or 100%. One and a half wholes is $100\% + 50\% = 150\%$.

Question 6

SELECTED RESPONSE — CORRECT: B

What percent is 9 out of 100?

- ☐ A 0.9%
- ☒ B 9%
- ☐ C 19%
- ☐ D 90%

A percent already compares to 100, so 9 out of 100 is exactly 9%.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: D

A robotics club had a fundraising goal of \$500. By Friday, they collected \$750. What percent of their goal did they achieve?

- ☐ A 50%
- ☐ B 75%
- ☐ C 125%
- ☒ D 150%

\$750 is 1.5 times the whole goal of \$500. $1.5 \times 100 = 150\%$, which represents 150% of the goal.

- **A:** That is the amount collected BEYOND the goal, not the percent of the goal reached.
- **B:** That compares \$750 to \$1000. The goal is \$500, and \$750 is more than the whole goal.
- **C:** That would be \$625 out of \$500. Divide the amount collected by the goal: 750 divided by 500.

Part B — correct answer: D

Which explanation best supports your answer to Part A?

- ☐ A Subtracting \$500 from \$750 gives \$250, and 250 is 75% of 500.
- ☐ B Since \$500 is half of \$1,000, the percent must be 50%.
- ☐ C The ratio 500 to 750 simplifies to 2 to 3, which is 125%.
- ☒ D Dividing \$750 by \$500 gives 1.5, and multiplying 1.5 by 100 gives 150%, representing 1.5 times the original whole.

A percent compares a part to a whole of 100. $750 / 500 = 1.5 = 150/100 = 150\%$.

Question 8

SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that correctly represent a rate per 100 or compare percents to one whole:

- ☒ **A** 100% represents exactly one whole quantity.
- ☒ **B** 45 out of 100 squares shaded on a 10×10 grid represents 45%.
- ☒ **C** A percent greater than 100% represents an amount that is more than the original whole.
- ☐ **D** 0.5% is equal to 50 out of 100.
- ☒ **E** A percent is a ratio that compares a number to 100.

Statements A, B, C, and E are true. Statement D is false because 0.5% is 0.5 out of 100 (1 out of 200), whereas 50 out of 100 is 50%.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A car dealership sold 50 cars last month: 20 sedans, 15 SUVs, 10 trucks, and 5 vans. The owner will order 100 cars for next month and wants the order to match last month's sales pattern. Write each type as a percent, then say how many of each to order — and explain why percents make this easier than the raw counts.

Model answer: Each count is out of 50, so multiply both terms by 2 to compare to 100: sedans $20/50 = 40/100 = 40\%$, SUVs $15/50 = 30/100 = 30\%$, trucks $10/50 = 20/100 = 20\%$, vans $5/50 = 10/100 = 10\%$. The percents total 100%, which confirms the whole is accounted for. Because the new order is exactly 100 cars, each percent IS the number to order: 40 sedans, 30 SUVs, 20 trucks, and 10 vans. Percents make this easier because they restate every count against the same whole of 100, so two months with different totals can be compared directly.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.